



Jeremías Likerman

ADJUNCT RESEARCHER, CONICET · ASSOCIATE PROFESSOR, UNIVERSITY OF BUENOS AIRES
Institute of Andean Studies "Don Pablo Groeber" (IDEAN), CONICET--UBA, Argentina
✉ jlikerman@uba.ar | 📧 Google Scholar | ORCID 0000-0003-3844-2085

Education

PhD, 2015	Geological Sciences, University of Buenos Aires (UBA)	Argentina
Thesis: <i>Prediction of fracturing on irregular geological surfaces using static methods</i> . Supervisor: Prof. Ernesto Cristallini.		
BSc, 2010	Geological Sciences, University of Buenos Aires (UBA)	Argentina

Main Career History

May 2026–present	Associate Professor, Department of Geological Sciences, UBA	Argentina
Teaching: Geostatistics and Geoinformatics.		
Sep 2025–present	Institutional Coordinator, IDEAN, CONICET-UBA	Argentina
2018–present	Adjunct Researcher (Investigador Adjunto), CONICET-IDEAN	Argentina
Structural geology, geomechanics, geodynamics, natural fractures, and geothermal systems. Research career from 2018; promoted to Investigador Adjunto, November 2023.		
May–Sep 2025	Visiting Researcher, GEO3BCN-CSIC	Spain
Aug–Sep 2024	Visiting Researcher, GEO3BCN-CSIC	Spain
2015–2017	Postdoctoral Researcher, CONICET and Universitat Politècnica de Catalunya	Argentina/Spain
2013–2026	Teaching appointments, Department of Geological Sciences, UBA	Argentina
2013–2014	Visiting Doctoral Researcher, Stanford University	USA

Major Research Achievements

- More than 15 years of research in structural geology and multiscale numerical modelling, integrating field observations, analogue experiments, geomechanics, and 2D/3D simulations.
- Developed predictive approaches for natural-fracture distribution and connectivity in folded structures and energy reservoirs, including the Yacoraite carbonates and the Vaca Muerta Formation.
- Principal Investigator of the 2026 Spanish supercomputing project on three-dimensional stress transfer between adjacent Andean flat slabs, and Co-PI of a 2026–2029 CONICET project linking tectonic-structural modelling with energy resources.
- Leads and co-leads national and international projects on subduction dynamics, lithospheric deformation, natural fractures, and geothermal systems; coordinator of CONICET–CSIC cooperation on geothermal modelling (2024–2026).
- Supervises early-career researchers and undergraduate, doctoral, and postdoctoral work in structural geology, geomechanics, and numerical modelling.

Selected Recent Major Publications

1. Jara, P.P., Pinto, L., & **Likerman, J.** (2026). Analogue modelling to enhance spatial reasoning and understanding of tectonic processes in geoscience education. *Andean Geology*, 53(3).
2. Villarroel, M.A., et al., including **Likerman, J.** (2026). The role of compressional structures in the development of collapse calderas: Insights from analogue models. *Journal of Volcanology and Geothermal Research*, 473, 108582.
3. Gianni, G.M., Gallo, L.C., **Likerman, J.**, et al. (2025). Slab underthrusting is the primary control on flat-slab size. *Science Advances*, 11(28).
4. Correa-Luna, C., Yagupsky, D.L., **Likerman, J.**, & Barcelona, H. (2025). Impact of large-scale structures on fracture network connectivity: Insights into the Vaca Muerta unconventional play. *Tectonophysics*, 230640.
5. Plotek, B., **Likerman, J.**, & Cristallini, E. (2024). Geomechanical modeling of fault-propagation folds: A comparative analysis of finite-element and the trishear kinematic model. *Journal of Structural Geology*, 105064.
6. Navarrete, C., et al., including **Likerman, J.** (2024). Massive Jurassic slab break-off revealed by a multidisciplinary reappraisal of the Chon Aike silicic large igneous province. *Earth-Science Reviews*, 249, 104651.

Full Curriculum Vitae

Professional Profile

Research profile: Geoscientist specializing in structural geology, geomechanics, and multiscale numerical modelling, with more than 15 years of experience in research, university teaching, fieldwork, and hands-on student training. His research addresses geodynamic, geothermal, and geomechanical processes in Andean systems and energy reservoirs, integrating two- and three-dimensional models to study subduction zones, crustal deformation, volcanic systems with geothermal potential, and natural and induced fractures, with emphasis on the Vaca Muerta Formation.

Teaching, outreach, and knowledge transfer: Teaching experience in Geological Surveying, Geostatistics, Geoinformatics, and Introduction to Geology, including practical work in geological mapping, spatial analysis, geological interpretation, geospatial tools, UAVs, and quantitative methods. Participation in public science communication, professional training, institutional management, and international scientific cooperation.

Academic and Postdoctoral Training

Postdoctoral Fellowship

Argentina/Spain

CONICET AND UPC

2015–2017

- Research topic: thermal evolution of oceanic lithosphere. Supervisors: Dr. Ernesto Cristallini (CONICET-UBA) and Dr. Sergio Zlotnik (UPC).

Visiting Researcher

USA

STANFORD UNIVERSITY

2013

- Courses in Structural Geology and MATLAB applied to Geosciences.

PhD in Geological Sciences

Argentina

UNIVERSITY OF BUENOS AIRES

2010–2015

- Thesis: *Prediction of fracturing on irregular geological surfaces using static methods*. Supervisor: Dr. Ernesto Cristallini.

BSc in Geological Sciences

Argentina

UNIVERSITY OF BUENOS AIRES

2005–2010

- Thesis: *Geology and structure of the Tres Cruces compound anticline, Jujuy, Argentina*. Supervisor: Dr. Ernesto Cristallini.

Appointments and Professional Experience

CURRENT APPOINTMENTS

Adjunct Researcher (Investigador Adjunto)

Argentina

CONICET-IDEAN

2018–present

- Research areas: fracture prediction, geodynamic modelling, and geothermal-potential modelling. CONICET research career from 2018; promoted to Investigador Adjunto, November 2023.

Associate Professor

Argentina

DEPARTMENT OF GEOLOGICAL SCIENCES, UNIVERSITY OF BUENOS AIRES

May 2026–present

- Courses: Geostatistics and Geoinformatics.

Institutional Coordinator

Argentina

IDEAN, CONICET-UBA

Sep 2025–present

- Council Member since April 2024.

PREVIOUS TEACHING APPOINTMENT

Teaching Assistant

Argentina

DEPARTMENT OF GEOLOGICAL SCIENCES, UNIVERSITY OF BUENOS AIRES

2013–2026

- Courses: Introduction to Geology, Geological Surveying, and Geostatistics.

PROFESSIONAL COURSES AND FIELD ACTIVITIES

Geology of Unconventional Reservoirs

Argentina

INSTRUCTOR IN CHARGE

2025

- Professional training course focused on the geological characterization of shale reservoirs.

Advanced Research Field Trip

Argentina

FIELD INSTRUCTOR

2025

- Organizer and instructor for *Perspectives on Vaca Muerta: subsurface attributes analysed from surface exposures*.

RESEARCH VISITS

Visiting Researcher

GEO3BCN-CSIC

Spain
May-Sep 2025

- Thermomechanical modelling of volcanic zones for geothermal-potential assessment. Host: Dr Ivone Jiménez-Munt.

Visiting Researcher

GEO3BCN-CSIC

Spain
Aug-Sep 2024

- Development of coupled numerical models for volcanic and geothermal systems.

Visiting Researcher

BARCELONATECH (UPC)

Spain
2015, 2017, 2019, 2022

- Research stays at the Numerical Computation Laboratory.

Research Projects

2026	Three-dimensional models of stress transfer between adjacent Andean flat slabs (Principal Investigator). AECT-2026-1-0080	Spain
2026-2029	Tectonic-structural analysis through analogue and numerical modelling: contributions to energy resources (Co-PI). PIP 11220250100817CO	Argentina
2025	Geothermal potential of volcanic zones (Principal Investigator). AECT-2025-1-0009	Spain
2025-2026	Structural controls in volcanic systems: magmatic intrusions and damage-zone development associated with hydrocarbon generation (Co-PI). Williams Foundation	Argentina
2024-2026	Assessment of geothermal potential in volcanic zones using thermomechanical numerical models (PI). LINCGL2024	Spain
2024	Geodynamic modelling of subduction zones and mantle plumes (PI). AECT-2024-2-0027	Spain
2023	Geodynamics of the Adria microplate, from mantle to surface (Co-PI). GEO3BCN	Spain
2022	Geodynamic modelling of subduction zones (PI). AECT-2022-3-0023	Spain
2020	Landscape dynamics and geological hazards associated with dams in mountain regions (Co-PI). PICT-2019-04011	Argentina
2020	Analogue and numerical modelling: contributions to tectonic and structural problems (Co-PI). PICT-2019-00997	Argentina
2018	Natural fracturing and first-order structures: predictive models applied to Cerro Domuyo (Co-PI). UBACyT	Argentina
2017	Geological evolution of the Andes and its economic and environmental impact (Co-PI). PUE 22920160100051	Argentina
2016	Numerical modelling of the thermal evolution of oceanic lithosphere (PI). PICT 2015-1226	Argentina
2016	Application of analogue and numerical models to tectonic-structural problems (Co-PI). PICT-2016-1407	Argentina

Research Supervision

RESEARCHER SUPERVISION

Plotek, Berenice

Argentina
Since Aug. 2025

ASSISTANT RESEARCHER

- Supervisor: Likerman, J.; co-supervisor: Cristallini, E.

POSTDOCTORAL SUPERVISION

Plotek, Berenice

Argentina
2024

FAULT-PROPAGATION FOLDS: A MECHANICAL APPROACH USING FINITE-ELEMENT NUMERICAL MODELLING

- Supervisors: Cristallini, E. and Likerman, J.

DOCTORAL SUPERVISION

Correa Luna, Clara

Argentina
2023

NATURAL FRACTURING IN THE VACA MUERTA FORMATION: CHARACTERIZATION AND MODELLING

- Supervisors: Yagupsky, D. and Likerman, J.

Undergraduate Thesis Supervision

Chandía Sepúlveda, Alexander Ignacio	Argentina
TECTONIC INVERSION OF A METAMORPHIC COMPLEX IN THE DESEADO MASSIF: INSIGHTS FROM ANALOGUE MODELLING	In progress
Supervisors: Likerman, J. and Navarrete, C.	
De Rosa, Tomás	Argentina
QUATERNARY DEFORMATION IN THE FIAMBALÁ VALLEY, CATAMARCA	In progress
Supervisors: Likerman, J. and Sagripanti, L.	
Blanco, Lucía	Argentina
GEOHERMAL-ELECTRIC POTENTIAL OF THE CERRO BLANCO VOLCANIC COMPLEX: INFLUENCE OF RHEOLOGY AND STRUCTURE THROUGH NUMERICAL MODELS	In progress
Supervisors: Likerman, J. and Lamberti, C.	
Celis Santisteban, Celso Angel	Argentina
STRUCTURAL EVOLUTION OF AN INVERTED METAMORPHIC COMPLEX IN THE DESEADO MASSIF: INSIGHTS FROM NUMERICAL MODELLING	In progress
Supervisors: Likerman, J. and Navarrete, C.	
Ruiz, Luciana	Argentina
ASSESSMENT OF THE GEOHERMAL-ELECTRIC POTENTIAL OF THE CERRO BLANCO VOLCANIC COMPLEX USING NUMERICAL MODELLING	2026
Supervisors: Lamberti, C. and Likerman, J.	
Gramajo, Pilar	Argentina
QUATERNARY DEFORMATION IN THE CHASCHUIL VALLEY, WESTERN SIERRA DE LAS PLANCHADAS, CATAMARCA	2025
Supervisors: Likerman, J. and Sagripanti, L.	
Acosta, Luana	Argentina
GEOMORPHOLOGY OF THE RÍO GRANDE VALLEY BETWEEN BARDAS BLANCAS AND PORTEZUELO DEL VIENTO, MENDOZA	2023
Supervisors: Winocur, D. and Likerman, J.	
Mazzini, Martín	Argentina
GEOLOGY OF THE EASTERN SLOPE OF CERRO PERITO MORENO, EL BOLSÓN, RÍO NEGRO	2022
Supervisors: Likerman, J. and Tobal, J.	
Relañez, Gastón	Argentina
STRUCTURE AND NEOTECTONICS OF THE TABLADITAS AREA, JUJUY, ARGENTINA	2019
Supervisors: Yagupsky, D. and Likerman, J.	
Ventisky, Esteban	Argentina
GEOLOGY, STRUCTURE, AND FRACTURING OF THE LAJAS FORMATION IN THE COVUNCO ANTICLINE, NEUQUÉN	2017
Supervisors: Likerman, J. and Tobal, J.	
Correa Luna, Clara	Argentina
STRUCTURAL GEOLOGY AND FRACTURING IN THE TRES CRUCES BASIN, JUJUY	2015
Supervisors: Yagupsky, D. and Likerman, J.	

Peer-Reviewed Publications

1. Analogue modelling to enhance spatial reasoning and understanding of tectonic processes in geoscience education. Jara, P. P., Pinto, L., **Likerman, J.** *Andean Geology* 53.3 (2026)

2. The role of compressional structures in the development of collapse calderas: Insights from analogue models. Villarroel, M. A., Browning, J., Jara, P. P., Harnett, C. E., Holohan, E. P., Marquardt, C. J., Guzman, M., **Likerman, J.** *Journal of Volcanology and Geothermal Research* (2026) p. 108582

3. Impact of large-scale structures on fracture network connectivity: Insights into the Vaca Muerta unconventional play, Neuquén basin, Argentina. Correa-Luna, C., Yagupsky, D. L., **Likerman, J.**, Barcelona, H. *Tectonophysics* (2025) p. 230640

4. Slab underthrusting is the primary control on flat slab size. Gianni, G. M., Gallo, L. C., **Likerman, J.**, Echaurren, A., Gianni, C. R., Faccena, C. *Science Advances* 11.28 (2025)

5. Massive Jurassic slab break-off revealed by a multidisciplinary reappraisal of the Chon Aike silicic large igneous province. Navarrete, C., Gianni, G., Tassara, S., Zaffarana, C., **Likerman, J.**, Márquez, M., Wostbrock, J., Planavsky, N., Tardani, D., Frasette, M. P. *Earth-Science Reviews* 249 (2024) p. 104651

6. Geomechanical modeling of fault-propagation folds: A comparative analysis of finite-element and the trishear kinematic model. Plotek, B., **Likerman, J.**, Cristallini, E. *Journal of Structural Geology* (2024) p. 105064

7. Ghost-arc geochemical anomaly at a spreading ridge caused by supersized flat subduction. Gianni, G. M., **Likerman, J.**, Navarrete, C. R., Gianni, C. R., Zlotnik, S. *Nature communications* 14.1 (2023) p. 2083

8. Natural fracture characterization of the Los Catutos Member (Vaca Muerta Formation) in the southern sector of the Sierra de la Vaca Muerta, Neuquén Basin, Argentina. Correa-Luna, C., Yagupsky, D. L., **Likerman, J.**, Barcelona, H. *Journal of South American Earth Sciences* 120 (2022) p. 104081

9. Kinematics of fault-propagation folding: Analysis of velocity fields in numerical modeling simulations. Plotek, B., Heckenbach, E., Brune, S., Cristallini, E., **Likerman, J.** *Journal of Structural Geology* 162 (2022) p. 104703

10. The influence of climate on the dynamics of mountain building within the Northern Patagonian Andes. García Morabito, E., Beltrán-Triviño, A., Terrizzano, C. M., Bechis, F., **Likerman, J.**, Von Quadt, A., Ramos, V. A. *Tectonics* 40.2 (2021) e2020TC006374

11. The effects of small-scale convection in the shallow lithosphere of the North Atlantic. **Likerman, J.**, Zlotnik, S., Li, C.-F. *Geophysical Journal International* 227.3 (2021) pp. 1512–1522

12. Fracture network analysis of Yacoraite Formation in the Tres Cruces sub-basin, northwestern Argentina. Luna, C. C., Yagupsky, D. L., **Likerman, J.** *Journal of South American Earth Sciences* (2019) p. 102446

13. Cenozoic intraplate tectonics in Central Patagonia: Record of main Andean phases in a weak upper plate. Gianni, G. M., Echaurren, A., Folguera, A., **Likerman, J.**, Encinas, A., García, H. P. A., Dal Molin, C., Valencia, V. A. *Tectonophysics* 721 (2017) pp. 151–166

14. Closure type effects on the structural pattern of an inverted extensional basin of variable width: Results from analogue models. Jara, P., **Likerman, J.**, Charrier, R., Herrera, S., Pinto, L., Villarroel, M., Winocur, D. *Journal of South American Earth Sciences* (2017)

15. Climatic and Tectonic forcing on alluvial fans in the Southern Central Andes. Terrizzano, C. M., Morabito, E. G., Christl, M., **Likerman, J.**, Tobal, J., Yamin, M., Zech, R. *Quaternary science reviews* 172 (2017) pp. 131–141

16. Structural and statistical characterization of joints and multi-scale faults in an alternating sandstone and shale turbidite sequence at the Santa Susana Field Laboratory: Implications for their effects on groundwater flow and contaminant transport. Cilona, A., Aydin, A., **Likerman, J.**, Parker, B., Cherry, J. *Journal of Structural Geology* 85 (2016) pp. 95–114

17. Middle to late Miocene extensional collapse of the North Patagonian Andes (41° 30'–42° S). Tobal, J. E., Folguera, A., **Likerman, J.**, Naipauer, M., Sellés, D., Boedo, F. L., Ramos, V. A., Gimenez, M. *Tectonophysics* 657 (2015) pp. 155–171

18. Geodynamic context for the deposition of coarse-grained deep-water axial channel systems in the Patagonian Andes. Ghiglione, M. C., **Likerman, J.**, Barberón, V., Giambiagi, L. B., Aguirre-Urreta, B., Suarez, F. *Basin Research* 26.6 (2014) pp. 726–745

19. Role of basin width variation in tectonic inversion: insight from analogue modelling and implications for the tectonic inversion of the Abanico Basin, 32°–34° S, Central Andes. Jara, P., **Likerman, J.**, Winocur, D., Ghiglione, M. C., Cristallini, E. O., Pinto, L., Charrier, R. *Geological Society, London, Special Publications* 399 (2014) SP399–7

20. Along-strike structural variations in the Southern Patagonian Andes: Insights from physical modeling. **Likerman, J.**, Burlando, J. F., Cristallini, E. O., Ghiglione, M. C. *Tectonophysics* 590 (2013) pp. 106–120

Fellowships and Distinctions

2027	Research Fellowship in Japan Kobe University	JSPS
2025	SZNet 2025 Chile Field Trip National Science Foundation	SZ4D
2022	Academic Mobility Fellowship among Partner Institutions Ibero-American University Association for Postgraduate Studies	AUIP
2016	International Research Stay National Scientific and Technical Research Council	CONICET
2015	Postdoctoral Fellowship National Scientific and Technical Research Council	CONICET
2013	Doctoral Research Stay at Stanford University Fulbright – Bunge & Born Foundation	Doctoral fellowship
2013	Doctoral Fellowship Type II National Scientific and Technical Research Council	CONICET
2010	Doctoral Fellowship Type I National Scientific and Technical Research Council	CONICET

Skills

Programming and reproducibility: $\LaTeX$, Python, R, MATLAB, HTML5, Git, and reproducible scientific documentation.

Modelling and HPC: Underworld2, LaMEM, GOLEM, ASPECT; analogue modelling, experimental design, PIV, post-processing, MPI, and high-performance-computing workflows.

Spatial analysis: QGIS, GMT, PostGIS; seismic interpretation, geological mapping, and UAV-assisted field methods.

Fieldwork: more than 15 years of experience, extensive knowledge of the Andes, campaign planning and logistics, and team leadership.

Languages: Spanish (native), English (fluent, C2), Italian (A2), Catalan (A2).